

**SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)**  
**Department of Computer Science and Engineering**

|                       |                                                                                                                     |                      |                                         |
|-----------------------|---------------------------------------------------------------------------------------------------------------------|----------------------|-----------------------------------------|
| <b>Course Code:</b>   | CSA10 / CSA1063                                                                                                     | <b>Course Title:</b> | Software Engineering                    |
| <b>CO Assessed:</b>   | CO1                                                                                                                 | <b>Assessment:</b>   | Assessment Tool 1 – Case Study Analysis |
| <b>Weightage:</b>     | 20%                                                                                                                 | <b>Total Marks:</b>  | 25                                      |
| <b>CO1 Statement:</b> | <i>Apply advanced methodologies and agile project management techniques to manage software projects effectively</i> |                      |                                         |
| <b>Student Name:</b>  | Dhanushree S                                                                                                        | <b>Reg. No.:</b>     | 192421218                               |
| <b>Date:</b>          | 22-07-2026                                                                                                          | <b>Duration:</b>     | 60 minutes                              |

**CO1 ASSESSMENT TOOL 1 – Case Study Analysis**

**Code of Conduct**

*I **Dhanushree S / 192421218** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: Dhanushree S

# Ai-Based Crop Disease Detection and Farmer Advisory Platform

## 1. Understand the Problem Statement

### ○ Explain the problem in your own words.

Agriculture is one of the most important sectors that supports food production and the economy. Crop diseases significantly reduce agricultural productivity and cause financial losses to farmers every year. Traditional disease detection depends on manual observation and agricultural experts, which is time-consuming and sometimes inaccurate. Many farmers are unable to identify diseases during the early stages, resulting in the spread of infection and reduced crop yield.

The proposed AI-Based Crop Disease Detection and Farmer Advisory Platform uses Artificial Intelligence and Image Processing techniques to automatically detect crop diseases from leaf images. The system provides accurate disease identification, treatment suggestions, fertilizer recommendations, and preventive measures. It also helps farmers monitor crop health, reduce crop damage, and make better farming decisions.

### Key Points

- Uses AI to identify crop diseases automatically.
- Detects diseases at an early stage.
- Provides treatment and prevention recommendations.
- Reduces crop damage and farming losses.
- Improves crop productivity and quality.
- Supports smart farming practices.

### ○ Identify the objectives of the proposed system.

The main objective of this project is to develop an intelligent system that helps farmers identify crop diseases quickly and accurately. The system aims to improve agricultural productivity while reducing crop losses.

## Objectives

- Detect crop diseases using Artificial Intelligence and Image Processing.
- Identify diseases during the early stages of infection.
- Provide suitable pesticide and fertilizer recommendations.
- Suggest preventive measures for healthy crop growth.
- Reduce crop losses and increase agricultural productivity.
- Improve the quality of crops through timely treatment.
- Reduce farmers' dependence on agricultural experts.
- Help farmers make faster and more informed decisions.
- Create a digital database of crop diseases for future analysis.
- Promote sustainable and modern farming practices.

### ○ **State the scope of the problem.**

The scope of the proposed system is to provide an AI-based solution for detecting crop diseases and supporting farmers in disease management. The system focuses on improving farming efficiency through technology.

## Scope

- Detect diseases in crop leaves using AI.
- Support different types of crops and diseases.
- Provide treatment and prevention recommendations.
- Help farmers monitor crop health continuously.
- Reduce crop losses caused by delayed disease detection.
- Improve agricultural productivity and quality.
- Integrate weather information for better farming decisions.
- Support future expansion with IoT sensors and drone technology.
- Provide multilingual support for farmers from different regions.
- Improve overall agricultural management using digital technology.

## 2. **Identify Key Stakeholders and Challenges**

- **Identify all the stakeholders involved in the system.**

The proposed system involves several stakeholders who contribute to the development, maintenance, and effective use of the platform.

#### Stakeholders

- Farmers
- Agricultural Experts
- System Administrators
- AI Developers
- Software Developers
- Government and Agricultural Organizations
- Researchers
- Technology Providers
- Database Administrators

- **Explain the roles and responsibilities of each stakeholder.**

#### **Farmers**

- Register and log in to the system.
- Upload crop leaf images.
- Receive disease detection reports.
- Follow treatment recommendations.
- Monitor crop health regularly.

#### **Agricultural Experts**

- Verify disease information.
- Recommend suitable pesticides and fertilizers.
- Provide preventive farming guidance.
- Improve advisory recommendations.

### **AI Developers**

- Develop AI models.
- Train the disease detection model.
- Improve prediction accuracy.
- Update AI algorithms regularly.
- Software Developers
- Design and develop the application.
- Integrate AI modules.
- Fix software bugs.
- Add new features.

### **System Administrators**

- Manage users and system data.
- Monitor system performance.
- Maintain security.
- Backup databases regularly.

### **Government and Agricultural Organizations**

- Provide farming policies.
- Conduct awareness programs.
- Share agricultural information.
- Promote smart farming.

### **Researchers**

- Collect disease datasets.
- Improve AI accuracy.
- Study emerging crop diseases.
- Technology Providers
- Supply cloud storage.
- Provide AI software tools.
- Maintain technical infrastructure.

- **Discuss the major challenges and issues faced by the stakeholders.**

### **Farmers**

- Limited digital literacy.
- Poor internet connectivity.
- Difficulty capturing clear crop images.
- Lack of smartphones in some areas.

### **Agricultural Experts**

- Need to verify AI recommendations.
- Different diseases require region-specific solutions.

### **AI Developers**

- Need large and high-quality datasets.
- Continuous model training.
- Handling new disease categories.

### **System Administrators**

- Maintaining server performance.
- Protecting user information.
- Managing increasing numbers of users.

### **Government Organizations**

- Promoting digital agriculture.
- Training farmers.
- Supporting rural internet infrastructure.

## **3. Analyze the Current Approach**

- **Describe the existing system or current process.**

### **Describe the existing system**

Currently, farmers mainly detect crop diseases through manual observation. They inspect leaves and plants visually to identify symptoms. If they cannot recognize the disease, they consult agricultural experts or nearby agricultural centers. The diagnosis depends heavily on the farmer's experience and expert availability, making the process slow and less efficient.

### **Existing Process**

- Farmers inspect crops manually.
- Observe leaf color and disease symptoms.
- Consult agricultural experts.
- Receive treatment recommendations.
- Purchase pesticides.
- Apply treatment.
- Monitor crop recovery.

### **Strengths of the Existing System**

- Easy to understand.
- No advanced technology required.
- Relies on expert knowledge.
- Low initial investment.
- Suitable for experienced farmers.

- **Identify its strengths and limitations**

### **Strengths**

- Detects crop diseases quickly using AI.
- Provides accurate disease identification.

- Saves farmers' time and effort.
- Reduces crop losses through early detection.
- Offers instant farming advice.
- Accessible through mobile devices.

### **Limitations**

- Accuracy depends on image quality.
  - Requires internet connection.
  - May not detect new or rare diseases.
  - Farmers need basic smartphone knowledge.
  - AI model requires regular updates.
- .
- **Analyze the problems or gaps that need to be addressed.**
    - Delay in identifying crop diseases.
    - Lack of expert guidance in rural areas.
    - Poor internet connectivity.
    - Limited support for local languages.
    - Inaccurate detection under poor lighting conditions.
    - Limited integration with weather and soil information.

## **4. Propose a Suitable Solution Using Software Engineering Principles**

- **Design a software-based solution for the given problem.**

The proposed AI-Based Crop Disease Detection and Farmer Advisory Platform is a smart agricultural application designed to help farmers identify crop diseases automatically using Artificial Intelligence. Farmers can upload crop leaf images through a mobile or web application. The AI model processes the image using deep learning algorithms and predicts the disease with a confidence score.



Based on the prediction, the system provides disease information, pesticide recommendations, fertilizer suggestions, organic treatment methods, and preventive measures. Additionally, the system offers weather forecasts, seasonal farming tips, and disease alerts to improve crop management. The platform stores all disease records securely and allows administrators to update disease information and AI models regularly.

- **Describe the major functional modules of the proposed system.**

### **1. User Management Module**

- Farmer registration and login
- Profile management
- Authentication and authorization

### **2. Crop Image Upload Module**

- Capture or upload crop leaf images
- Validate image quality
- Store uploaded images

### **3. AI Disease Detection Module**

- Process uploaded images
- Detect crop diseases using AI
- Display disease name and confidence score

### **4. Farmer Advisory Module**

- Suggest suitable pesticides
- Recommend fertilizers
- Provide organic treatment methods
- Offer preventive measures

### **5. Weather & Notification Module**

- Display weather forecasts
- Send disease alerts
- Provide seasonal farming tips

## 6. Admin Module

- Manage users
- Update disease database
- Train and update AI models
- Monitor system performance

- **Identify the functional and non-functional requirements.**

### Functional Requirements

- User registration and secure login
- Upload crop images
- AI-based disease detection
- Display disease details
- Provide treatment and prevention recommendations
- Show weather information
- Send notifications and alerts
- Allow administrators to manage users and update the system

### Non-Functional Requirements

- **Performance:** Disease detection should be completed within a few seconds.
- **Reliability:** The system should provide accurate and consistent results.
- **Scalability:** Support a growing number of users, crops, and diseases.
- **Security:** Protect user information through secure authentication and data encryption.
- **Usability:** Provide a simple and user-friendly interface for farmers.
- **Maintainability:** Allow easy updates to the AI model and disease database.
- **Availability:** Ensure the system is accessible whenever farmers need it.

- **Explain how software engineering principles such as modularity, scalability, maintainability, reliability, usability, and security are incorporated into your solution.**

### **Modularity**

The system is divided into independent modules such as User Management, Image Upload, AI Detection, Farmer Advisory, Weather, and Admin. This makes the software easier to develop, test, and maintain.

### **Scalability**

The platform is designed to support more users, additional crop varieties, and new disease detection models without affecting existing functionality.

### **Maintainability**

The AI model, disease database, and advisory information can be updated independently, making the system easier to maintain and improve.

### **Reliability**

The AI model is trained using quality datasets to provide accurate disease detection and dependable recommendations. Error handling ensures stable system performance.

### **Usability**

The application features a simple interface with easy navigation, clear instructions, and support for regional languages, making it accessible to farmers with limited technical knowledge.

### **Security**

The platform uses secure user authentication, encrypted data storage, role-based access control, and regular backups to protect user information and maintain system integrity.

## 5. Justify the Chosen Methodology/Framework

- Select an appropriate Software Development Life Cycle (SDLC) model or software development framework (e.g., Agile, Scrum, Waterfall, Spiral, DevOps).

### Agile Scrum

Agile Scrum is selected for developing the **AI-Based Crop Disease Detection and Farmer Advisory Platform** because it supports iterative development, continuous testing, and regular feedback. Since AI models and user requirements may change over time, Agile Scrum provides the flexibility to improve the system in every sprint.

- **Justify why the selected methodology/framework is suitable for the proposed system.**

**Agile Scrum** is the most suitable software development methodology for the AI-Based Crop Disease Detection and Farmer Advisory Platform because it supports continuous development, flexibility, and regular improvements throughout the project lifecycle. Since agricultural requirements, AI technologies, and user expectations may change over time, Agile allows the development team to respond quickly to these changes without affecting the overall project.

The project is divided into smaller iterations called sprints, where each sprint delivers a working part of the system. This enables developers to identify problems early, receive user feedback, and improve the application continuously. The methodology also promotes teamwork, regular communication, and quality assurance, making it highly suitable for developing intelligent agricultural applications.

### Flexible Requirements

The requirements of farmers, agricultural experts, and AI technologies may change over time. New crop diseases may appear, treatment methods may be updated, and farmers may request additional features. Agile Scrum allows these changes to be incorporated into the

project without restarting the entire development process. This flexibility ensures that the software remains useful and relevant to the changing needs of the agricultural sector.

## Incremental Development

Agile divides the entire project into smaller development phases known as sprints. Each sprint focuses on developing one or more modules such as User Management, Image Upload, AI Disease Detection, Farmer Advisory, Weather Information, or Admin Management. This approach allows the team to deliver a working system step by step, making development easier and reducing project risks. Farmers and stakeholders can start using completed features while the remaining modules are still under development.

## Continuous Feedback

One of the major advantages of Agile Scrum is continuous feedback from stakeholders. After every sprint, the developed features are demonstrated to farmers, agricultural experts, and project stakeholders. Their suggestions and feedback help improve system performance, user interface, disease detection accuracy, and advisory recommendations. Continuous feedback ensures that the final product satisfies the real needs of end users.

## Faster Bug Fixes

Since testing is performed after every sprint, software defects are identified much earlier than in traditional development methods. Developers can immediately correct coding errors, improve system functionality, and update AI models before moving to the next sprint. This reduces the overall maintenance effort and improves software quality.

## Improved AI Model

The AI disease detection model requires regular training with new crop images and updated datasets. Agile Scrum allows developers to continuously retrain and improve the machine learning model after every sprint. As more disease images become available, the system accuracy increases, enabling better disease prediction and treatment recommendations.

## Better Team Collaboration

Agile Scrum encourages collaboration among software developers, AI engineers, agricultural experts, testers, database administrators, project managers, and stakeholders. Daily meetings, sprint planning, sprint reviews, and retrospective sessions help every team member understand project progress and responsibilities. Better communication leads to improved software quality and timely project completion.

## Early Delivery of Software

Instead of waiting until the entire project is completed, Agile delivers functional modules at regular intervals. Farmers can start using available features such as user registration or disease detection while developers continue implementing additional functionalities. This approach increases customer satisfaction and reduces project delivery time.

## Improved Software Quality

Continuous integration, regular testing, code reviews, and stakeholder feedback ensure that software quality remains high throughout development. Errors are detected and corrected immediately, reducing the possibility of major failures after deployment.

- **Explain how it supports successful project development and maintenance.**

Agile Scrum supports successful project development by promoting continuous planning, regular communication, teamwork, and iterative improvements. Since the AI-Based Crop Disease Detection and Farmer Advisory Platform requires frequent updates to disease databases, AI models, weather services, and treatment recommendations, Agile provides the flexibility needed for long-term maintenance. Every sprint improves the software based on user feedback, making the application more reliable, accurate, and user-friendly.

## Delivers Working Software in Short Development Cycles

The project is divided into short sprints lasting two to four weeks. At the end of every sprint, a working version of the software is delivered. This enables early testing, faster deployment, and immediate user feedback.

## Encourages Regular Communication

Daily Scrum meetings, sprint reviews, and planning sessions improve communication among developers, AI engineers, agricultural experts, testers, and project managers. Regular discussions help identify issues quickly and improve collaboration.

## Continuous Testing and Quality Assurance

Testing activities are performed throughout every sprint rather than waiting until project completion. Unit testing, integration testing, system testing, and user acceptance testing help ensure that every module functions correctly.

## Supports Easy Addition of New Features

As agriculture evolves, new crops, diseases, weather conditions, and treatment methods may need to be added. Agile allows these new features to be incorporated into future sprints without affecting the existing system.

## Simplifies Software Maintenance

Since each module is developed independently, maintenance becomes easier. Developers can update the AI model, disease database, weather module, or advisory system separately without interrupting other modules.

## Continuous Improvement of AI Models

The disease detection model can be retrained using newly collected crop images. Continuous learning improves prediction accuracy and enables the system to detect newly emerging crop diseases more effectively.

## Better Risk Management

Regular sprint reviews help identify technical risks, software defects, and project delays at an early stage. Early risk identification reduces development costs and improves project success.

## High Customer Satisfaction

Continuous involvement of farmers and agricultural experts ensures that the developed software meets user expectations. Their suggestions directly influence future improvements, resulting in a more practical and user-friendly application.

## Reliable and Scalable Software

Agile ensures that the system remains reliable, scalable, secure, and maintainable throughout its lifecycle. New technologies and AI improvements can be integrated without redesigning the complete system.

## 5. Discuss Expected Outcomes and Limitations

The proposed **AI-Based Crop Disease Detection and Farmer Advisory Platform** offers significant benefits to farmers, agricultural experts, and the overall farming sector. By integrating Artificial Intelligence (AI), Image Processing, and modern software technologies, the system provides fast and accurate crop disease detection, enabling farmers to take timely action before diseases spread. This helps reduce crop losses, improve productivity, and support sustainable farming practices.

The system allows farmers to upload crop leaf images through a mobile or web application, where the AI model analyzes the image and identifies the disease. Based on the detected disease, the platform recommends suitable pesticides, fertilizers, organic treatments, and preventive measures. It also provides weather updates and farming tips, helping farmers make informed decisions regarding crop management.



## **Expected Benefits**

- Early detection of crop diseases reduces crop damage and prevents the spread of infection.
- AI-based disease identification provides fast and accurate diagnosis.
- Timely treatment recommendations improve crop health and reduce yield loss.
- Proper pesticide and fertilizer suggestions minimize unnecessary chemical usage.
- Reduces farming costs by preventing excessive pesticide application.
- Improves crop productivity and enhances the quality of agricultural products.
- Provides instant expert guidance through a mobile or web application.
- Saves farmers' time and effort compared to manual disease identification.
- Weather forecasts and farming alerts help farmers protect crops from environmental risks.
- Digital storage of crop and disease records supports future monitoring and analysis.
- Reduces dependence on agricultural experts, especially in remote areas.
- Encourages the adoption of modern digital farming technologies.
- Promotes sustainable and environmentally friendly farming practices.
- Increases farmers' income by reducing crop losses and improving production.

### **○ Describe the expected benefits of the proposed solution.**

The proposed AI-Based Crop Disease Detection and Farmer Advisory Platform is designed to improve the efficiency of crop disease management by using Artificial Intelligence (AI) and Image Processing technologies. The system helps farmers detect crop diseases quickly and accurately, enabling them to take timely action before the disease spreads. By providing instant disease identification, treatment recommendations, and preventive measures, the platform reduces crop losses and improves overall agricultural productivity.

The application allows farmers to upload crop leaf images through a mobile or web interface. The AI model analyzes the image, identifies the disease, and provides detailed recommendations, including suitable pesticides, fertilizers, organic treatment methods, and preventive measures. The system also integrates weather information and farming alerts, helping farmers make informed decisions based on environmental conditions. This

improves crop health, reduces farming costs, and supports sustainable agricultural practices.

Overall, the proposed solution enhances farming efficiency, minimizes manual effort, improves disease management, and contributes to higher crop yield and better-quality agricultural products.

#### Expected Benefits

- Detects crop diseases at an early stage, reducing the spread of infection and minimizing crop damage.
- Provides fast and accurate disease identification using Artificial Intelligence and Image Processing techniques.
- Recommends suitable pesticides, fertilizers, and organic treatment methods based on the detected disease.
- Helps farmers take timely preventive measures, improving crop health and reducing production losses.
- Increases agricultural productivity by ensuring healthy crop growth and better harvest quality.
- Reduces dependence on agricultural experts by providing expert guidance through the application.
- Saves farmers' time and effort by automating the disease detection process.
- Reduces farming costs by minimizing unnecessary pesticide and fertilizer usage.
- Provides weather forecasts and farming alerts to support better crop management decisions.
- Stores crop disease records digitally, enabling future monitoring and analysis.
- Offers a simple and user-friendly interface that can be easily used by farmers with basic smartphone knowledge.
- Supports sustainable farming by encouraging responsible pesticide usage and environmentally friendly agricultural practices.
- Improves the accuracy of disease detection through continuous AI model training and database updates.
- Increases farmers' income by reducing crop losses and improving overall farm productivity.

- Promotes the adoption of modern digital agriculture technologies, contributing to long-term agricultural development and food security.
- **Explain how the solution addresses the identified challenges.**

The proposed system helps overcome the problems identified in the current system by:

- Detecting crop diseases at an early stage using AI image analysis.
  - Providing instant treatment and prevention recommendations.
  - Offering weather-based farming advice to improve crop management.
  - Reducing the time required for disease diagnosis.
  - Making expert guidance available anytime through the application.
  - Supporting better farming decisions with accurate and reliable information.
- 
- **Discuss potential risks, implementation challenges, and limitations.**

### **Potential Risks**

- Incorrect AI predictions may lead to improper treatment recommendations.
- Data privacy and security risks if user information is not protected.
- System failures or server downtime may affect availability.

### **Implementation Challenges**

- Collecting a large and high-quality dataset for AI model training.
- Ensuring good image quality for accurate disease detection.
- Providing stable internet connectivity in rural areas.
- Training farmers to use the application effectively.

### **Limitations**

- AI accuracy depends on the quality of uploaded images.
- The system may not detect rare or newly emerging crop diseases.

- Continuous updates are required to improve the AI model.
- Limited functionality in areas with poor internet connectivity.

- **Suggest possible future enhancements.**

The following enhancements can improve the system in the future:

- Develop an offline mode for disease detection.
- Add voice assistance in regional languages.
- Integrate IoT sensors for soil moisture and temperature monitoring.
- Use drones for automatic crop monitoring.
- Provide market price information for crops.
- Predict diseases using weather and climate data.
- Include an AI chatbot for instant farmer support.
- Expand the platform to support more crops and diseases.
- Integrate government schemes and subsidy information for farmers.